INCH-POUND

A-A-59440/2C
06 December 2018
SUPERSEDING
A-A-59440/2B
21 May 2012

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

# COCK-VALVE, DRAIN, WITH HOSE BIB, 150 PSI, NON-FLAMMABLE LIQUID OR GAS, NPTF-1 THREADS

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring the cock-valve, drain, with hose bib, 150 psi, non-flammable liquid or gas, NPTF-1 threads described herein shall consist of this document and the issue in effect of A-A-59440.

CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the PIN as shown in the following example (see NOTES).

| AA59440 | /02 | - | 002 | Blank or C |
|---|---|---|---|---|
| CID number | Applicable CID spec sheet (two digits) | | Dash number (see figure 1) | Handle plating (see table I and figure 1) |

SALIENT CHARACTERISTICS.

Interface and physical dimensions. Cock-valves, drain, with hose bib, 150 psi, non-flammable liquid or gas, NPTF-1 threads supplied to this CID specification sheet shall be as specified herein (see figure 1).

Sizes. Nominal sizes of 1/8 inch through 1/4 inch with NPTF-1 threads are covered by this CID.

Intended applications. These items are for use in non-flammable liquid or gas systems where pressures do not exceed 150 psig (10.34 bar) at 150°F (66°C).

Permissible variations in dimensions. Figure 1 lists the permissible variation in dimensions for each size. The external dimensions of the cock-valve should not limit procurement unless the end item applications restrict these dimensions.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:14Z
Check the source to verify that this is the current version before use.

Threads. Threads are dryseal NPTF-1 in accordance with A-A-59440.

Materials.

Brass. Brass shall be in accordance with the following commercial standards:

| | | | |
|---|---|---|---|
| ASTM B16/B16M | ASTM B121/B121M | ASTM B427 | SAE AMS4845 |
| ASTM B22 | ASTM B124/B124M | ASTM B505/B505M | SAE AMS4855 |
| ASTM B30 | ASTM B148 | ASTM B584 | SAE AMS4860 |
| ASTM B36/B36M | ASTM B176 | ASTM B763 | SAE AMS4862 |
| ASTM B61 | ASTM B271/B271M | ASTM B770 | SAE AMS4890 |
| ASTM B62 | ASTM B369 | ASTM B806 | |

Carbon steel. Cast, forged, or machined carbon steel shall be in accordance with QQ-S-698.

Handle. Handle may be hot rolled or cold rolled steel conforming to QQ-S-698 or another commonly accepted industry standard.

Protective coating:

TABLE I. Material and finish identification codes.

| PIN code material/plating finish | Material | Plating finish |
|---|---|---|
| Blank | Steel or Brass | Brass with no plating or steel with Zinc Plating. (See Note 6/ and Table II) 2/ 6/ |
| B | Brass | No additional plating. 2/ |
| C | Carbon Steel | Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 2. 1/ 5/ |
| Z | Carbon Steel | Zinc in accordance with ASTM B633; type V or VI, Fe/Zn 8. 4/ |
| J | Carbon Steel | Zinc-nickel in accordance with SAE-AMS2417, 12% to 16% nickel, type 2, grade B. (315 ¼ µ inches (8¼ µm) minimum) 3/ |
| H | Carbon Steel | Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 3/ 5/ |

1/ Embrittlement test need not be run. Cadmium shall not be used in oxygen or potable water systems. Cadmium is prohibited from use in NAVSEA owned shipboard systems.
2/ Brass shall not have any lead contents.
3/ Hexavalent chromium free. Finish shall be RoHS compliant.
4/ For use in low corrosion environments.
5/ For use in medium to high corrosion environments.
6/ These parts were originally listed in MS35782 as a single part type.

Test pressures:

a. Shell: 150 psig (10.3 bar).
b. Seat: 50 psig (3.45 bar).

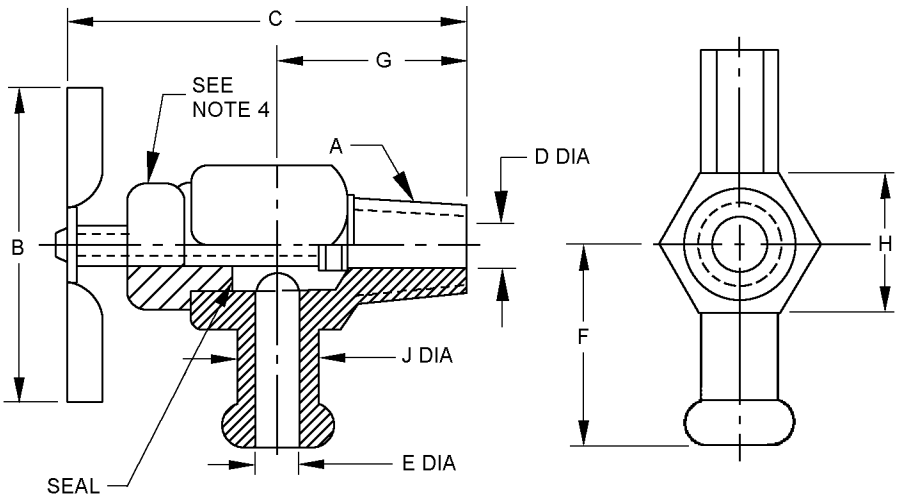
Valve bodies. Valve bodies shall be provided with hexagonal wrenching surfaces.

Valve stem. Drain valves shall be designed to prevent the valve stem from being inadvertently removed.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:14Z
Check the source to verify that this is the current version before use.

Optional designs. Figure 1 is for identification purposes only and is not intended to restrict designs or shapes not dimensioned.




| Dash No. | Handle plating | A pipe thread NPTF-1 | B max (mm) | C open max (mm) | D diameter min | E diameter +.03 -.06 (+0.8 mm) (-1.5 mm) | F max (mm) | G max (mm) | H ±.015 (0.38 mm) | J diameter max (mm) |
|---|---|---|---|---|---|---|---|---|---|---|
| 009 | Zinc or Brass | 1/8-27 | 1.38 (35.1) | 1.88 (47.8) | .19 (4.8) | .25 (6.4) | .88 (22.4) | .78 (19.8) | .56 or .62 (14.2 or 15.7) | .34 (8.6) |
| 009B | Brass | | | | | | | | | |
| 009C | Cadmium | | | | | | | | | |
| 009Z | Zinc | | | | | | | | | |
| 009J | Zinc-nickel | | | | | | | | | |
| 009H | Zinc-Aluminum | | | | | | | | | |
| 010 | Zinc or Brass | 1/4-18 | 1.38 (35.1) | 2.00 (50.8) | .22 (5.6) | .25 (6.4) | .88 (22.4) | .78 (19.8) | .56 or .62 (14.2 or 15.7) | .34 (8.6) |
| 010B | Brass | | | | | | | | | |
| 010C | Cadmium | | | | | | | | | |
| 010Z | Zinc | | | | | | | | | |
| 010J | Zinc-nickel | | | | | | | | | |
| 010H | Zinc-Aluminum | | | | | | | | | |

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.010 inch (0.25 mm).
4. Diameter of hole shall be adequate to allow drainage through the valve.
5. Thread “A” shall be a modified NPTF-1, the effective NPTF-1 thread is shortened by one thread on the small end.

FIGURE 1. Dimensions and configuration cock-valve, drain, with hose bib.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:14Z
Check the source to verify that this is the current version before use.

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents

FEDERAL SPECIFICATION

QQ-S-698 - Steel, Sheet and Strip, Low Carbon

COMMERCIAL ITEM DESCRIPTION

A-A-59440 - Cock-Valves and Valves, General Requirements for

(Copies of these documents are available online at http://quicksearch.dla.mil or https://assist.dla.mil).

Other Publications

ASTM INTERNATIONAL

| | | |
|---|---|---|
| ASTM B16/B16M | - | Free-Cutting Brass Rod, Bar and Shapes for use in Screw Machines |
| ASTM B22 | - | Bronze Castings for Bridges and Turntables |
| ASTM B30 | - | Copper Alloys in Ingot Form |
| ASTM B36/B36M | - | Brass Plate, Sheet, Strip, and Rolled Bar |
| ASTM B61 | - | Steam or Valve Bronze Castings |
| ASTM B62 | - | Composition Bronze or Ounce Metal Castings |
| ASTM B121/B121M | - | Leaded Brass Plate, Sheet, Strip, and Rolled Bar |
| ASTM B124/B124M | - | Copper and Copper Alloy Forging Rod, Bar, and Shapes |
| ASTM B148 | - | Aluminum-Bronze Sand Castings |
| ASTM B176 | - | Copper-Alloy Die Castings |
| ASTM B271/B271M | - | Copper-Base Alloy Centrifugal Castings |
| ASTM B369 | - | Copper-Nickel Alloy Castings |
| ASTM B427 | - | Gear Bronze Alloy Castings |
| ASTM B505/B505M | - | Copper Alloy Continuous Castings |
| ASTM B584 | - | Copper Alloy Sand Castings for General Applications |
| ASTM B633 | - | Electrodeposited Coatings of Zinc on Iron and Steel |
| ASTM B763 | - | Copper Alloy Sand Castings for Valve Applications |
| ASTM B770 | - | Copper-Beryllium Alloy Sand Castings for General Applications |
| ASTM B806 | - | Copper Alloy Permanent Mold Castings for General Applications |

(Copies of these documents are available online at http://www.astm.org or from the ASTM International, P.O. Box C700, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:14Z
Check the source to verify that this is the current version before use.

SAE INTERNATIONAL

| | | |
|---|---|---|
| SAE-AMS-QQ-P-416 | - | Plating, Cadmium (Electrodeposited) |
| SAE AMS4845 | - | Tin Bronze Castings, Sand and Centrifugal<br>87.5CU - 10SN - 2ZN as Cast |
| SAE AMS4855 | - | Leaded Red Brass, Sand and Centrifugal Castings<br>85CU - 5.0SN - 5.0PB - 5.0ZN as Cast |
| SAE AMS4860 | - | Manganese Bronze, Sand and Centrifugal Castings<br>58CU - 39ZN - 1.2FE - 1.0AL - 0.80MN as Cast |
| SAE AMS4862 | - | Manganese Bronze, Sand and Centrifugal Castings<br>63CU - 24ZN - 6.2AL - 3.8MN - 3.0FE High Strength, as Cast |
| SAE AMS4890 | - | Copper-Beryllium Alloy Castings<br>97CU - 2.1BE - 0.52CO - 0.28SI Solution Heat Treated (TBOO) |

(Copies of these documents are available on line at www.sae.org from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, and Tel: 877-606-7323 [inside USA and Canada] or 724-776-4970 [outside USA], email at CustomerService@sae.org.)

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

| MFR's CAGE | MFR's name and address |
|---|---|
| 01276 | Eaton Aeroquip Inc.<br>14615 Lone Oak Rd.<br>Eden Prairie, MN 55344<br>952-294-2032/888-258-0222<br>jeffenoe@eaton.com |
| 93061 | Parker-Hannifin Corp<br>Brass Products Division<br>300 Parker Drive<br>Otsego, MI 49708-1472<br>260-636-2104<br>dwsanders@Parker.com |

Military standard (MS) supersession data. This CID replaces the following MS as shown in table I.

TABLE I. MS supersession data.

| CID PIN 1/ | Superseded MS |
|---|---|
| AA59440/1-009 | MS35783-1 |
| AA59440/2-009C | MS35783-1 |
| AA59440/2-010 | MS35783-2 |
| AA59440/2-010C | MS35783-2 |

1/ These parts were originally listed in MS35783 as a single part type. Due to special handling/disposal and corrosion resistance requirements the parts have been separated in this CID.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:14Z
Check the source to verify that this is the current version before use.

Part number P/N supersession data. These CID specification sheet PINs supersede the following MFR's P/N's as shown in table II. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. P/N supersession data.

| Dash number (see table I) AA59440/1 | MFR's CAGE | MFR's P/N 1/ |
|---|---|---|
| 009 | 01276 | 131-16667 |
| 009C | - | - |
| 010 | 01276 | 131-16680 |
| 010C | - | - |

1/ The manufacturer's P/N shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see marking paragraph of A-A-59440.

National stock number (NSN). The following is a list of NSN's assigned which correspond to this CID specification sheet. The list in table III is for information only and may not be indicative of all possible NSN's associated with the CID specification sheet. For up to date information on assigned NSN's, please contact the aforementioned DLA Land and Maritime office listed in A-A-59440.

TABLE III. NSN's.

| Dash number (see figure 1) AA59440/1 | NSN |
|---|---|
| 010 | 4820-00-845-1096 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:14Z
Check the source to verify that this is the current version before use.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

MILITARY INTERESTS:

Custodians:
Army - AR
Navy - SH
Air Force - 99
DLA - CC

Review activities:
Army - AT, CR4, MI
Navy– AS, SA

CIVIL AGENCY AND COORDINATING ACTIVITIES:

GSA - FSS

Preparing activity

DLA - CC

(Project 4820-2018-011)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organization and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:14Z
Check the source to verify that this is the current version before use.